Two unit cubes in a cube in every dimension: reductions checked in Lean, routes that fail, and the evidence for $s_d(2) = 2$

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Abstract

Let $s_d(n)$ be the side of the smallest d-cube that holds n unit d-cubes with disjoint interiors, each placed by an arbitrary rigid motion. It is known that $s_d(2) = 2$ for $d \le 6$: Paper I of this series proves it in Lean for $d \le 5$, and Hosono proved $d = 6$ with computer assistance. We know of no result for $d \ge 7$, and no statement of the problem in the literature before Hosono's. This paper collects what we know about the conjecture that $s_d(2) = 2$ in every dimension. *Equivalent forms.* The conjecture is Hosono's Lemma F restricted to orthogonal matrices whose rows have ℓ^1 norm at most 2. It is also a statement about the largest $(d-1)$-cube in a simplex whose vertices lie on orthogonal axes. *Proofs in Lean, in every dimension.* The near-axis step, an off-column step and a pivot inequality reduce the induction on d to an explicit remaining case, and give $d \le 4$ without any computation. A new aligned step loses nothing. With it, the conjecture in every dimension follows from one geometric statement, the *facet lemma*: some largest cube in such a simplex has a facet parallel to a facet of the simplex. *The Helmert conjecture.* A recursive face-on-face placement should give the exact largest cube; this would imply the facet lemma. It contains the classical square in a triangle and Croft's theorem on the largest cube in a regular tetrahedron. For the regular n-simplex it predicts the side $1/(\sqrt{2}\sum_{k=2}^{n+1}\sqrt{1-1/k})$. *Routes that fail.* We record, with counterexamples verified to 50–60 digits, the failure of every single-pivot induction ($d = 5, 6, 9$), of the positive-part strengthening ($d = 8$), of convex relaxations, among them Hosono's spectral bound ($d \ge 6$), and of local arguments for the facet lemma ($d = 4$). *Evidence.* A Hosono-style numerical search in dimensions 3 to 32 finds no counterexample. Away from the coordinate axes, and in the remaining case of the induction, F stays far below 1. Conjecture H and the facet lemma hold in every test up to $d = 12$.

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1 Introduction

1.1 The problem

In dimension d write $s_d(n)$ for the smallest side of a cube that holds n unit d-cubes with disjoint interiors, each placed by an arbitrary rigid motion. The grid of 2^d cubes gives $s_d(n) \le 2$ for $n \le 2^d$, and the volume bound gives $s_d(n) \ge n^{1/d}$. If two unit cubes already need side 2, then $s_d(n) = 2$ for all $2 \le n \le 2^d$ (Paper I [2], plateau theorem).

Conjecture 1.1 (Two cubes). $s_d(2) = 2$ for every $d \ge 1$. Equivalently, $s_d(n) = 2$ for every $d \ge 1$ and every $2 \le n \le 2^d$.

Table 1 gives the status. Hosono [5] reduced the conjecture to an inequality for orthogonal matrices, Lemma F (Section 2). He proved it for $d \le 6$ with computer assistance, and Paper I [2] proved it in Lean for $d \le 5$. Hosono states the problem for general d in his issue #21 [6]. He reports no proof of $s_d(2) = 2$ for any $d \ge 3$ in the literature, and Friedman's survey [7] and his page on cubes in cubes give no lower bound for two cubes. We searched for the other forms below (frames, the largest cube in a simplex, ℓ^1-minimal orthonormal bases) and found no earlier

d	status of $s_d(2) = 2$	source
$1, 2$	true	trivial; for squares see Friedman [7]
$3, 4$	true, in Lean	Paper I [2]; here a third proof without computation (Corollary 3.3)
5	true, in Lean	Paper I [2] (1174 interval cases checked by Lean)
6	true, computer-assisted	Hosono [5]: 2.8 million simplex certificates; not in Lean, not peer reviewed
≥ 7	open	here: reduced to the facet lemma (Theorem 3.11); no counterexample in $d \leq 32$ (Section 6)

Table 1: The conjecture $s_d(2) = 2$ by dimension.

statement. Two special cases of the geometric form are classical, and we use them as checks (Section 4.4).

1.2 Results

Every statement below carries a label. [Lean] means checked in Lean 4 with Mathlib, using only the standard axioms, and linked to its declaration. [proof] means proved in this paper on paper. [numerical] means a numerical observation, with the script that produced it. [literature] means quoted from the literature.

1. **Equivalent forms** (Section 2). The conjecture is equivalent to restricted Lemma F, and to a statement about the largest cube in a simplex whose vertices lie on orthogonal axes. Further forms use frames, chords and mixed volumes [proof, Lean for the identities].
2. **Reductions in every dimension** (Section 3) [Lean]. Hosono's near-axis step holds for the restricted lemma in every dimension. With an elementary bound it gives $s_d(n) = 2$ for $d \leq 4$ without computation. An off-column step and a pivot inequality reduce the induction from $d - 1$ to d to an explicit remaining case. An aligned step, new here, gives $F_j \leq 1$ exactly when a row has at most one further nonzero entry.
3. **Main Theorem B** (Theorem 3.11) [Lean]. If the facet lemma holds in every dimension $d \geq 6$, then $s_d(n) = 2$ for $2 \leq n \leq 2^d$ in every dimension.
4. **The Helmert conjecture** (Section 4). The recursive face-on-face cube should be the largest cube in a simplex with vertices on orthogonal axes. This implies the facet lemma. The facet lemma holds for $d = 3$ [proof]. Conjecture H holds for the regular tetrahedron by Croft's theorem [literature]. For regular simplices it predicts an explicit side (Conjecture 4.7).
5. **Routes that fail** (Section 5). Each comes with the reason it fails and, where possible, a verified counterexample.
6. **Numerical evidence** in dimensions 3 to 32 (Section 6) and a summary of all the evidence (Section 7).

Figure 1 shows how the statements fit together.

1.3 How to read the Lean links

Every [Lean] statement is followed by its declarations, linked to the source at commit `a58da7c247` of the repository [1]. The files of this paper are `LemmaFr.lean`, `OffColumn.lean`, `PivotStep.lean` and `FacetStep.lean`, on top of the files of Paper I. Section 8 lists them. The scripts and data of Sections 5 and 6 are in `scripts/optimality/lemmaF_all_d/` at the same repository; Section 8 gives the commit.

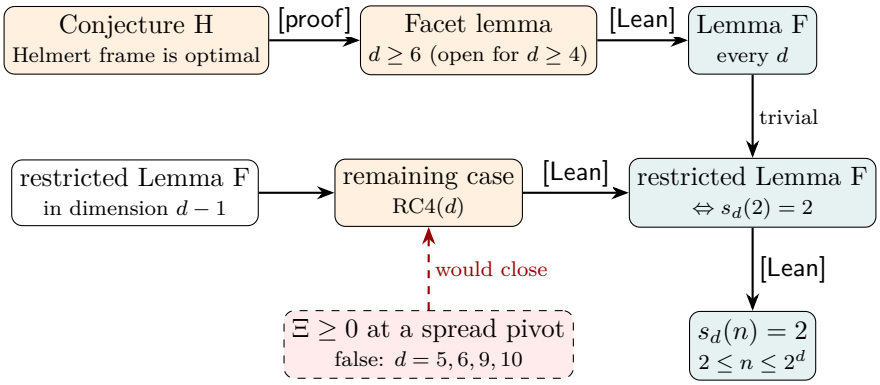


Figure 1: The reductions. Green: proved (in Lean unless marked). Orange: open. Red, dashed: a route that fails. The vertical chain on the right holds in every dimension. The left branch is the pivot induction of Section 3; the top row is the route through the facet lemma.

2 The conjecture and its equivalent forms

This section. The conjecture as an inequality for orthogonal matrices, for frames, and as a statement about the largest cube in a simplex with vertices on orthogonal axes.
Lean. `CentreGeneral.lean`, `FacetStep.lean` (`Fcol_eq_colValue`).

2.1 Lemma F

For an orthogonal $d \times d$ matrix R write $w_i = \|R_i\|_1$ for the ℓ^1 norm of row i and, for a column j,

$$F_j(R) = \sum_{i=1}^{d} |R_{ij}|\,(2 - w_i) \qquad (\texttt{Fcol}).$$

Lemma F in dimension d (`LemmaF`) is the statement $F_j(R) \leq 1$ for every orthogonal R and every j. *Restricted Lemma F* (`LemmaFr`) is the same statement for the orthogonal matrices whose rows all have $w_i \leq 2$. For the identity $F_j = 1$. For a rotation in one coordinate plane, with $\sigma = |\cos\theta| + |\sin\theta|$, $F_j = 1 - (\sigma - 1)^2$.

The rows of a unit cube placed in a cube of side $s < 2$ have ℓ^1 norm at most s, because w_i is the cube's extent in coordinate i. So only the restricted lemma is needed.

Theorem 2.1 (Hosono; Paper I [Lean])**.** *If restricted Lemma F holds in dimension d, then every unit d-cube inside a cube of side $s < 2$ contains the centre in its interior, and $s_d(n) = 2$ for every $2 \leq n \leq 2^d$.*

Lean: `centre_of_lemmaF_rows`, `cube_plateau_of_centre_lemma`.

The converse holds for the rows of norm strictly below 2.

Proposition 2.2 ([proof])**.** *If $s_d(2) = 2$, then $F_j(R) \leq 1$ for every orthogonal R whose rows all have $w_i < 2$.*

Proof. Let $w_i < 2$ for all i and $F_j(R) > 1$. Choose $s < 2$ with $\max_i w_i < s$ and $\sum_i |R_{ij}|(s - w_i) > 1$. Place the unit cube $g(Q_d)$, $g(y) = Ry + b$, so that its centre m satisfies $m_i - \frac{s}{2} = -\operatorname{sgn}(R_{ij})\frac{s-w_i}{2}$. This is admissible [2, §3]. For the container's centre $c = \frac{s}{2}\mathbf{1}$ the cube coordinate $y = R^\top(c - b)$ has $y_j - \frac{1}{2} = \sum_i R_{ij}(c_i - m_i) = \sum_i |R_{ij}|\frac{s-w_i}{2} > \frac{1}{2}$, so $c \notin \operatorname{int} g(Q_d)$. A hyperplane through c then separates $\operatorname{int} g(Q_d)$ from c. The point reflection of $g(Q_d)$ in c lies in the same container, and the two cubes have disjoint interiors (`interior_disjoint_of_separated`). So two unit cubes fit in side $s < 2$. □

The induction below needs the closed condition $w_i \le 2$, because the polar factor of a minor (Section 3) satisfies only that. Proposition 2.2 and Theorem 2.1 differ only on the boundary $\max_i w_i = 2$.

2.2 The value of a column, and frames

Fix j and write $x_i = |R_{ij}|$, $S = \|x\|_1$. The *value* of column j is

$$V_j(R) = \sum_i x_i\,(w_i - x_i) = \sum_i x_i\,\|\text{row } i \text{ without column } j\|_1 \qquad (\texttt{colValue}).$$

Lemma 2.3 ([Lean]). *$F_j(R) = 2S - 1 - V_j(R)$ for orthogonal R. So $F_j(R) \le 1$ if and only if $V_j(R) \ge 2(S-1)$.*

Lean: `Fcol_eq_colValue`.

Proof. $x_i(2 - w_i) = 2x_i - x_i(w_i - x_i) - x_i^2$, and $\sum_i x_i^2 = 1$. □

Changing the signs of rows does not change F_j, V_j or orthogonality, so we may take column j equal to $v = x \ge 0$. The other columns then form an orthonormal basis of $v^\perp$, a *frame*: a $d \times (d-1)$ matrix C with $C^\top C = I$ and $C^\top v = 0$. Its rows C_i have Gram matrix $I - vv^\top$. Every frame arises this way, and

$$\mathrm{Val}_v(C) = \sum_i v_i\,\|C_i\|_1 = V_j(R).$$

Corollary 2.4 (frame form). *Lemma F in dimension d holds if and only if $\mathrm{Val}_v(C) \ge 2(\|v\|_1 - 1)$ for every unit $v \ge 0$ in $\mathbb{R}^d$ and every frame C of $v^\perp$. Restricted Lemma F is the same statement for the frames with $v_i + \|C_i\|_1 \le 2$ for all i.*

2.3 The largest cube in a simplex with vertices on orthogonal axes

For a unit $v > 0$ and $\lambda > 0$ let

$$\Delta_\lambda(v) = \{\xi \in \mathbb{R}^d :\ \xi \ge 0,\ \langle v, \xi\rangle = \lambda\},$$

the $(d-1)$-simplex with vertices $(\lambda/v_i)e_i$. A unit $(d-1)$-cube in its hyperplane is $\{\xi_0 + \sum_k t_k c_k : t \in [0,1]^{d-1}\}$, with edge directions the columns c_k of a frame C of $v^\perp$.

Theorem 2.5 ([proof]). *A unit $(d-1)$-cube with frame C fits in $\Delta_\lambda(v)$ by a translation if and only if $\lambda \ge \frac{1}{2}\mathrm{Val}_v(C)$. It fits in the relative interior if and only if $\lambda > \frac{1}{2}\mathrm{Val}_v(C)$.*

Proof. Over the cube, coordinate i ranges over $[\xi_{0i} - N_i,\ \xi_{0i} + P_i]$, where $N_i = \sum_k (C_{ik})^-$ and $P_i = \sum_k (C_{ik})^+$. So the cube lies in $\Delta_\lambda(v)$ if and only if $\xi_{0i} \ge N_i$ for all i and $\langle v, \xi_0\rangle = \lambda$. Such a ξ_0 exists if and only if $\lambda \ge \langle v, N\rangle$. Since $\sum_i v_i C_{ik} = 0$, the positive and negative parts of $v_i C_{ik}$ have equal sums, so $\langle v, N\rangle = \frac{1}{2}\sum_{i,k} v_i |C_{ik}| = \frac{1}{2}\mathrm{Val}_v(C)$. The strict version is the same with strict inequalities. □

By scaling, the largest cube in $\Delta_\lambda(v)$ has side $2\lambda/\min_C \mathrm{Val}_v(C)$, attained since the frames form a compact set. Write the vertices as $t_i e_i$, so $v_i = \lambda/t_i$, $\lambda = h = (\sum_i t_i^{-2})^{-1/2}$ is the distance from the origin to the simplex, and $\|v\|_1 = h\sum_i 1/t_i$. Corollary 2.4 becomes:

Corollary 2.6 ([proof]). *Lemma F in dimension d is equivalent to the following statement: for all $t_1, \dots, t_d > 0$, every $(d-1)$-cube inside the simplex with vertices $t_1 e_1, \dots, t_d e_d$ has side at most*

$$\frac{1}{\sum_i 1/t_i - 1/h}, \qquad h = \Big(\sum_i t_i^{-2}\Big)^{-1/2}.$$

The simplices with vertices on mutually orthogonal rays from a point are exactly the orthocentric simplices whose orthocentre lies inside: their squared edge lengths have the form $\lambda_a + \lambda_b$ with all $\lambda_a > 0$ [11]. In the plane these are the acute triangles. For the regular simplex (v uniform) Corollary 2.6 gives side at most $1/(\sqrt{2}(d - \sqrt{d}))$ for unit edge. The true values are smaller (Table 2). Numerically Lemma F is tight only at axis-aligned columns, so the bound of Corollary 2.6 is reached only in the limit of a degenerate simplex.

Remark 2.7 (the restricted lemma). The cubes of the restricted lemma are those whose frame rows satisfy $v_i + \|C_i\|_1 \le 2$. Geometrically: the cube's edge directions are not too far from the coordinate directions. We do not use this form.

2.4 Chords and mixed volumes

Proposition 2.8 (chords; [proof]). *Let Δ be an n-simplex and $c_1, \dots, c_n$ orthonormal. A unit cube with edge directions c_k fits in Δ by a translation if and only if*

$$\sum_{k=1}^{n} \frac{1}{L_\Delta(c_k)} \le 1,$$

where $L_\Delta(c)$ is the length of the longest chord of Δ in direction c.

Proof. Let $\Delta = \{y : \langle u_a, y\rangle \le b_a\}$ with unit facet normals u_a and facet areas f_a, so $\sum_a f_a u_a = 0$ and $\sum_a f_a b_a = n \operatorname{vol}\Delta$ (origin inside). The cube $y_0 + \sum_k t_k c_k$ fits if and only if $\langle u_a, y_0\rangle \le b_a - \sum_k \langle u_a, c_k\rangle^+$ for all a. These are $n+1$ inequalities in $y_0 \in \mathbb{R}^n$ whose normals have the single linear relation $\sum_a f_a u_a = 0$. By Farkas' lemma they are solvable if and only if $\sum_a f_a \sum_k \langle u_a, c_k\rangle^+ \le n \operatorname{vol}\Delta$. For a simplex, $n \operatorname{vol}\Delta = L_\Delta(c) \sum_a f_a \langle u_a, c\rangle^+$: the right side is $L_\Delta(c)$ times the area of the projection of Δ along c, and the chord length is a cone function over that projection. □

The same computation says that a unit cube $\square$ fits if and only if the mixed volume satisfies $V(\Delta[n-1], \square) \le \operatorname{vol}\Delta$, since $nV(\Delta[n-1], \square) = \sum_a f_a h_\square(u_a)$. With $\Delta = \Delta_{S-1}(v)$ (so $n = d-1$), Lemma F says $\sum_k 1/L_k \ge 1$ for every frame, and also $V(\Delta[n-1], \square) \ge \operatorname{vol}\Delta$ for every unit cube $\square$. The script `forms_check.py` checks Theorem 2.5 and Proposition 2.8 against linear programs on 300 random instances each.

Proposition 2.9 (the average over the columns; [proof]). *For every orthogonal R, $\sum_j (1 - F_j(R)) = \sum_i (w_i - 1)^2$.*

Proof. $\sum_j F_j = \sum_i w_i(2 - w_i)$ and $d = \sum_i 1$. □

So Lemma F holds on average over the columns, with room $\sum_i (w_i - 1)^2$. The difficulty is that one column can take more than its share.

3 What is proved in every dimension

This section. Reductions from dimension $d-1$ to dimension d that hold in every dimension, all in Lean: the near-axis step, a bound for the remaining case, the off-column step, the pivot inequality and its concentrated case, and the aligned step. Consequences: $s_d(n) = 2$ for $d \le 4$ without computation; the induction reduced to an explicit remaining case; and every dimension reduced to the facet lemma (Theorem 3.11).
Lean. `LemmaFr.lean`, `OffColumn.lean`, `PivotStep.lean`, `FacetStep.lean`.

3.1 Block form and the polar factor

We use Hosono's block form and polar factor as in Paper I [2, §5]. Around an entry $c = R_{pq} > 0$,

$$R \simeq \begin{pmatrix} c & r^\top \\ x & M \end{pmatrix}, \qquad Q = M - \frac{1}{1+c}\, x\, r^\top$$

(`BlockOrth`, `polarQ`). Here Q is orthogonal when $x \neq 0$ (`polarQ_orth`), $Q^\top x = -r$ (`polarQ_transpose_x`), and the rows move by a rank-one term, $M_i = Q_i + \frac{1}{1+c} x_i\, r^\top$ in the convention of `polarQ_entry`. When the pivot is in column j ($q = j$), $F_j(R)$ is compared with F of all the columns of Q (the near step). When $q \neq j$, column j survives in Q with entries $Q_{ij} = M_{ij} - \frac{1}{1+c} x_i r_j$, and $F_j(R)$ is compared with $F_j(Q)$ (the off-column, pivot and aligned steps).

3.2 The near step and the bound for the remaining case

Theorem 3.1 (near step in every dimension; [Lean]). *Assume restricted Lemma F in dimension $d-1$. Let R be orthogonal of size d with all rows of ℓ^1 norm at most 2, and let $R_{i_0 j} \neq 0$ be an entry whose row has $\|R_{i_0}\|_1 - |R_{i_0 j}| \leq 1$. Then $F_j(R) \leq 1$.*

Lean: `near_step`, `step_at`; Hosono's `near_identity` and `lemmaK`.

Proof. Paper I proves the step for $d - 1 \leq 4$, where every unit row has ℓ^1 norm at most 2. In general the rows of Q satisfy $\|Q_i\|_1 \leq \|M_i\|_1 + \frac{1}{1+c}|x_i|\|r\|_1 \leq 2$, because $\|r\|_1 \leq 1 \leq 1 + c$ and the row i of R has norm at most 2 (`polar_row_le`, `coef_le_one`). So Lemma K applies to Q with the restricted hypothesis, and Hosono's identity gives $1 - F_j \geq K(Q, x) + 2(1 - c)(1 - \|r\|_1) \geq 0$. □

Theorem 3.2 (remaining bound; [Lean]). *If every row i with $R_{ij} \neq 0$ has $w_i - |R_{ij}| > 1$, then $F_j(R) \leq \|x\|_1 - 1$.*

Lean: `remaining_bound`.

Proof. Each term $x_i(2 - w_i) \leq x_i(1 - x_i)$, and $\sum_i x_i(1 - x_i) = S - 1$. □

Corollary 3.3 ($d \leq 4$ without computation; [Lean]). *Restricted Lemma F, the cube centre lemma, and $s_d(n) = 2$ for $2 \leq n \leq 2^d$ hold for every $d \leq 4$.*

Lean: `lemmaFr_le_four`, `centre_le_four`, `col_l1_le_two`.

Proof. Induction on d. If some entry of column j qualifies, use Theorem 3.1. Otherwise Theorem 3.2 gives $F_j \leq S - 1 \leq \sqrt{d} - 1 \leq 1$. □

This is the third proof for $d = 3, 4$, after Hosono's and Paper I's, and the shortest: it uses no case split and no certificate.

Theorem 3.4 (induction step; [Lean]). *Restricted Lemma F in dimension $d-1$, together with the statement $F_j(R) \leq 1$ for the matrices of the* remaining case

$$\mathrm{RC}(d): \quad w_i \leq 2\ \forall i, \qquad \|x\|_1 > 2, \qquad w_i - x_i > 1 \text{ whenever } x_i \neq 0,$$

gives restricted Lemma F in dimension d.

Lean: `RemainingCase`, `lemmaFr_succ`.

3.3 The off-column step

The extremal matrices of the remaining case include *sinks*: a row where column j is tiny and some other entry is large. For them we pivot outside column j.

Theorem 3.5 (off-column step; [Lean]). *Assume restricted Lemma F in dimension $d-1$. Let R be orthogonal with rows of ℓ^1 norm at most 2, let p be a row where $|R_{pj}|$ is smallest, and let $q \neq j$ be a column with $R_{pq}^2 \geq 3/4$. Then $F_j(R) \leq F_j(Q) \leq 1$.*

Lean: `offcol_step` (block form), `offcol_at`, `term_bound`; induction step `lemmaFr_succ2`.

Idea. A term-by-term comparison of $|M_{ij}|(2 - |x_i| - \|M_i\|_1)$ with $|Q_{ij}|(2 - \|Q_i\|_1)$, using the triangle inequality for the rows of Q. The error terms add up to $-c\,r_j$ by the orthogonality of columns q and j, $c\,r_j + \sum_i x_i M_{ij} = 0$. The bound $R_{pq}^2 \geq 3/4$ makes $\|r\|_1 \leq 1 + c$ and $\|r\|_2^2 \leq 1/4$, and the minimality of row p controls the signs. □

3.4 The pivot inequality

Normalise the signs so that column j is nonnegative and the pivot $c = R_{pq}$ is positive (`flipM`). Put $\nu = R_{pj}$, $y_i = M_{ij}$, $a = 1/(1+c)$, $L = \|r\|_1$, $\rho = L - \nu$ (the part of row p outside columns j and q), $\kappa_1 = 1 + c - \rho$ and $G_i = 2 - w_i$, and define

$$\Xi = \nu\big(c(c+L) - 1\big) + \sum_i \Big(2\kappa_1 \max(x_i, 0)\,(y_i - a\nu x_i) - \nu\, x_i G_i\Big) \qquad (\texttt{Xi}).$$

Theorem 3.6 (pivot inequality; [Lean]). *If $a\nu x_i \leq y_i$ for all i, then $F_j(R) \leq F_j(Q) - a\,\Xi$. In particular, with restricted Lemma F in dimension $d-1$, $\Xi \geq 0$ and an admissible pivot ($\|r\|_1 \leq 1 + c$) give $F_j(R) \leq 1$.*

Lean: `pivot_bound`, `pivot_step`, `pivot_at_gen`.

The only estimate in the proof is the triangle inequality for the rows of Q; the rest is the orthogonality of columns q and j and $\|\text{column } q\|_2 = 1$.

Theorem 3.7 (concentrated pivot; [Lean]). *Let p be a row where column j is smallest, and $c = |R_{pq}|$ the largest entry of row p, with $q \neq j$. If row p is concentrated, $\|R_p\|_1 - c - \nu \leq c + \nu/2$, then $\Xi \geq 0$, so restricted Lemma F in dimension $d-1$ gives $F_j(R) \leq 1$. For $d \geq 5$ the largest entry of a smallest row lies outside column j.*

Lean: `pivot_caseA`, `caseA_ok`, `pivot_at`, `exists_pivot`.

Idea. Two facts. First, $c(c+L) \geq c^2 + \|r\|_2^2 = 1$, because c is the largest entry of a unit row. Second, every row with $x_i > 0$ has $cG_i \leq 2c - 2cx_i - (c+\nu)y_i$, from its orthogonality to row p. □

Theorem 3.8 (the induction with every reduction; [Lean]). *Restricted Lemma F in dimension $d-1$ gives restricted Lemma F in dimension d, provided $F_j(R) \leq 1$ for the matrices of the case* RC4(d)*: the matrices of* RC(d) *in which every pair (p a row where column j is smallest, $q \neq j$ a column holding the largest entry of row p) has a spread row, $\rho > c + \nu/2$, and $c^2 < 3/4$.*

Lean: `RemainingCase4`, `lemmaFr_succ4`.

Section 5.2 shows that $\Xi \geq 0$ fails at spread pivots, even inside RC4, so this branch needs a different argument there. Numerically F is far below 1 on all of RC(d) (Section 6.4).

3.5 The aligned step and the facet lemma

Call a row p *aligned* (for column j) if it has at most one nonzero entry outside column j. For the cube of Theorem 2.5, row p of the frame is aligned exactly when the facet $\{\xi_p = 0\}$ of the simplex is parallel to a pair of facets of the cube.

Theorem 3.9 (aligned step; [Lean])**.** *Assume Lemma F in dimension $d-1$. If row p of an orthogonal $d \times d$ matrix R has no nonzero entry outside the columns j and $q \neq j$, then*

$$F_j(R) \le 1 - 2(1-|R_{pj}|)(1-|R_{pq}|) \le 1.$$

Lean: `aligned_step` (block form), `aligned_at`, `Fcol_le_one_of_axis_row`.

Proof. If $R_{pq} = 0$, row p is $\pm e_j$ and $F_j(R) = 1$. Otherwise flip the sign of column q so that $c = R_{pq} > 0$, and take the block form at (p, q). Then $r = \nu e_j$ with $\nu = R_{pj}$, so $c^2 + \nu^2 = 1$ and $\|x\|_2^2 = 1 - c^2 = \nu^2$. Since $Q^\top x = -r = -\nu e_j$ and Q is orthogonal, $x = -\nu\, Q_{\cdot j}$. The polar factor changes only column j: $Q_{ik} = M_{ik}$ for $k \neq j$, and $M_{ij} = Q_{ij} + \frac{\nu}{1+c} x_i = (1 - \frac{\nu^2}{1+c}) Q_{ij} = c\, Q_{ij}$. So with $u_i = |Q_{ij}|$, a unit vector, the ℓ^1 norms are $\|M_i\|_1 = \|Q_i\|_1 - (1-c)u_i$, and

$$F_j(R) = |\nu|(2-c-|\nu|) + \sum_i c u_i (2 - |\nu| u_i - \|Q_i\|_1 + (1-c)u_i) = |\nu|(2-c-|\nu|) + c\, F_j(Q) + c(1-c-|\nu|).$$

With $F_j(Q) \le 1$ and $c^2 + \nu^2 = 1$ this is at most $2|\nu| + 2c - 1 - 2c|\nu| = 1 - 2(1-|\nu|)(1-c)$. □

Unlike the pivot inequality, this step loses nothing: when row p is aligned, the polar factor is the minor with column j rescaled, and the computation is an identity. The price is Lemma F in dimension $d-1$ without the restriction, because the rescaling can raise the row norms of Q above 2.

Definition 3.10 (facet lemma; `FacetLemma`)**.** The *facet lemma in dimension d*: for every orthogonal $d \times d$ matrix R and column j there is an orthogonal R' with $|R'_{ij}| = |R_{ij}|$ for all i, $V_j(R') \le V_j(R)$, and an aligned row. Equivalently, by Theorem 2.5 and compactness: for every unit $v \ge 0$, some largest cube in $\Delta_1(v)$ has a pair of facets parallel to a facet of the simplex.

Theorem 3.11 (Main Theorem B; [Lean])**.** *If the facet lemma holds in every dimension $d \ge 6$, then Lemma F holds in every dimension, every unit d-cube inside a cube of side $s < 2$ contains the centre, and $s_d(n) = 2$ for every $d \ge 1$ and every $2 \le n \le 2^d$.*

Lean: `lemmaF_succ_of_facet`, `lemmaF_of_facet_from`, `lemmaF_of_facet`, `centre_of_facet`, `cube_plateau_of_facet`.

Proof. Lemma F in dimension 5 is Paper I's `lemmaF5`. If Lemma F holds in dimension $d-1$ and the facet lemma in dimension d, take R' as in Definition 3.10. Theorem 3.9 gives $F_j(R') \le 1$, and by Lemma 2.3 $F_j(R) = 2S - 1 - V_j(R) \le 2S - 1 - V_j(R') = F_j(R')$. Dimensions $d \le 4$ are Corollary 3.3. □

The facet lemma is trivial for $d = 2$ (`facetLemma_two`) and holds for $d = 3$ (Corollary 4.5). It follows from Conjecture H (Proposition 4.3). It is open for $d \ge 4$; numerically it holds in every test (Section 6.5).

4 The Helmert conjecture and the facet lemma

This section. The recursive face-on-face cube and its value; the Helmert conjecture and what it implies; the structure of minimisers; the facet lemma for $d = 3$; regular simplices, and Croft's theorem as a check.

Scripts. `frames.py`, `forms_check.py`, `star.py`, `rules.py`, `flag.py`, `locmin.py`.

4.1 The Helmert frame

Sort $v_1 \geq v_2 \geq \cdots \geq v_d \geq 0$ and let $A_k = v_1^2 + \cdots + v_{k-1}^2$. The vectors

$$b_k \;\propto\; A_k\, e_k - v_k\,(v_1, \ldots, v_{k-1}, 0, \ldots, 0), \qquad k = 2, \ldots, d,$$

form a frame of $v^\perp$, the *Helmert frame* (for uniform v it is the classical Helmert basis of the sum-zero hyperplane). Its value is

$$H(v) = \mathrm{Val}_v(b_2, \ldots, b_d) = 2\sum_{k=2}^{d} v_k \sqrt{A_k / A_{k+1}}, \qquad A_{d+1} = 1.$$

Row d (the smallest entry of v) is aligned: it meets only b_d. Deleting it gives the Helmert frame of $v' = (v_1, \ldots, v_{d-1})/a$, $a = \sqrt{1 - v_d^2}$, and

$$H(v) = a\, H(v') + 2 v_d\, a. \tag{1}$$

Geometrically: the Helmert cube stands on the facet of the simplex opposite its farthest vertex, and the facet of the cube on top of it is again a Helmert cube in a smaller simplex, one dimension down.

Proposition 4.1 ([proof]). *$H(v) \geq 2(\|v\|_1 - 1)$, with equality only when v has a single nonzero entry.*

Proof. Put $\alpha = \sqrt{A_k}$, $\beta = v_k$, $r = \sqrt{\alpha^2 + \beta^2} = \sqrt{A_{k+1}}$. Then $\beta\alpha/r \geq \alpha + \beta - r$, because $r^2 - (\alpha+\beta)r + \alpha\beta = (r-\alpha)(r-\beta) \geq 0$. Summing over $k \geq 2$, $\frac{1}{2}H(v) \geq \sum_{k\geq 2} v_k - (\sqrt{A_{d+1}} - \sqrt{A_2}) = \|v\|_1 - 1$. Equality in every step needs $r = \alpha$ or $r = \beta$, so $v_k = 0$ or $A_k = 0$ for each k. □

So for every v that is not axis-aligned, the Helmert cube is strictly smaller than the bound of Corollary 2.6.

Conjecture 4.2 (H). For every unit $v \geq 0$, the Helmert frame minimises Val_v over all frames of $v^\perp$. Equivalently, the largest $(d-1)$-cube in $\Delta_1(v)$ has side $2/H(v)$ and is the recursive face-on-face cube.

Proposition 4.3 ([proof]). *Conjecture H in dimension d implies the facet lemma in dimension d. Hence Conjecture H in all dimensions $d \geq 6$ implies Conjecture 1.1.*

Proof. Given R and j, let R' have column j equal to $v = |\text{column } j|$ and the Helmert frame as its other columns. It is orthogonal, its row at the smallest entry of v is aligned, and $V_j(R') = H(v) \leq \mathrm{Val}_v(C) = V_j(R)$, where C is the frame of the row-normalised R. Then apply Theorem 3.11. □

Proposition 4.1 and Proposition 4.3 give the chain of Fig. 1: Conjecture H ⇒ facet lemma ⇒ Lemma F ⇒ restricted Lemma F ⇔ $s_d(2) = 2$.

4.2 Minimisers and the facet lemma in dimension 3

Proposition 4.4 ([proof]). *Rotate two frame vectors c_k, c_l by an angle θ in their plane. Then $\theta \mapsto \mathrm{Val}_v$ is a sum of functions $\rho_i(|\cos(\theta + \alpha_i)| + |\sin(\theta + \alpha_i)|)$, one for each row i. Each is concave between consecutive kinks, and the kinks are the angles where one of the two rotated entries of row i vanishes. So at a local minimum of Val_v either some row has a kink there, meaning exactly one of its two entries in columns k, l is zero, or Val_v is constant along the rotation up to the next kink.*

Proof. Write row i restricted to columns k, l as $\rho_i(\cos\alpha_i, \sin\alpha_i)$; the rotation turns it into $\rho_i(\cos(\theta+\alpha_i), \sin(\theta+\alpha_i))$, and the other entries do not change. The function $|\cos t| + |\sin t|$ is concave on each interval between consecutive multiples of $\pi/2$. A sum of concave functions has a local minimum inside an interval only if it is constant near that point. □

Corollary 4.5 (facet lemma for $d = 3$; [proof])**.** *For $d = 3$ some minimiser of Val_v has a zero entry, hence an aligned row (rows have only two entries). With Theorem 3.9 this proves Lemma F in dimension 3.*

Proof. For $d = 3$ the frames of $v^\perp$ are one rotation family (up to sign changes, which do not change Val_v). Its minimum is attained, and by Proposition 4.4 it can be moved to a kink. □

This is Paper I's three-dimensional proof [2, §4] read geometrically. For $d \geq 4$, Proposition 4.4 applies to every pair of columns and gives zero entries, but not a full aligned row. The local minima in Section 5.7 show that no local argument can give more.

4.3 The frame step

When no row is aligned, the aligned step has a lossy analogue. Remove row p ($\nu = v_p$, $a = \sqrt{1-\nu^2}$) by rotating $\mathbb{R}^{d-1}$ in the plane of C_p and e_q so that C_p becomes $a\,e_q$; then delete row p and column q. This gives a frame C'' for $v' = v_{\neq p}/a$.

Conjecture 4.6 (frame step; [numerical]). If $\|C_p\|_1 \leq a + 2|C_{pq}|$, then $\mathrm{Val}_v(C) \geq a\,\mathrm{Val}_{v'}(C'') + 2v_p|C_{pq}|$.

For an aligned row, $|C_{pq}| = a$ and this is the identity behind Theorem 3.9. A derivation sketch: write $C_i = s_i + \beta_i\hat{u}$ with $\hat{u} = C_p/a$ and $\beta_i = -\nu v_i/a$; the triangle inequality in each row gives a bound whose first-order term cancels by $\sum_i v_i s_{iq} = 0$. The script `star.py` checked it on 97,617 cases inside the domain, with smallest margin $6 \cdot 10^{-10}$; outside the domain it fails. With Lemma F for v' it closes the induction when $v_p(a - |C_{pq}|) \leq (1-a)(1-v_p)$. In an adversarial search (`rules.py`, $d = 5, 6, 8$), the rule "remove the row that maximises the right-hand side" always closed the induction. The rule "largest v_p" failed once, slightly, next to an axis-aligned v. The rules "smallest v_p" and "most concentrated row" fail, in line with Section 5.2.

A candidate global mechanism (`flag.py`): with v sorted decreasingly and $\beta_k(c) = \sum_{i\geq k}(\mathrm{sgn}(c_k)\,v_i c_i)^+$, numerically $\sum_j \max_{k\geq 2}\beta_k(c_j) \geq H(v)/2$ for every frame (c_j), with equality at the Helmert frame ($d = 3$–7). This is slightly stronger than Conjecture H, and not proved.

4.4 Regular simplices

For uniform v the simplex is regular, and Conjecture H becomes explicit. A regular n-simplex of unit edge is $\Delta_\lambda(v)$ with $d = n+1$, $v = \mathbf{1}/\sqrt{d}$ and $\lambda = 1/\sqrt{2d}$, and $H(v) = \frac{2}{\sqrt{d}}\sum_{k=2}^{d}\sqrt{1-1/k}$.

Conjecture 4.7 (the largest cube in a regular simplex). The largest n-cube in a regular n-simplex of unit edge has side

$$\sigma_n = \frac{1}{\sqrt{2}\,\sum_{k=2}^{n+1}\sqrt{1-1/k}},$$

attained by the recursive face-on-face cube. Lemma F gives only $\sigma_n \leq 1/(\sqrt{2}(n+1-\sqrt{n+1}))$.

Two cases are known.

- $n = 2$: $\sigma_2 = 2\sqrt{3} - 3 \approx 0.4641$, the largest square in an equilateral triangle, with a side on a side of the triangle. More generally, Corollary 4.5 and Conjecture 4.2 for $d = 3$ say that the largest square in an acute triangle has a side on a side of the triangle; Conway and Guy [10] discuss the placements of the largest square.

n	σ_n (Conjecture 4.7)	Lemma F bound	volume bound
1	1.000000	1.207107	1.000000
2	0.464102	0.557678	0.658037
3	0.295907	0.353553	0.490280
4	0.215315	0.255834	0.390664
5	0.168482	0.199156	0.324686
6	0.138033	0.162395	0.277772
7	0.116720	0.136730	0.242702
8	0.101001	0.117851	0.215494
9	0.088948	0.103413	0.193771
10	0.079422	0.092031	0.176027
11	0.071711	0.082839	0.161259
12	0.065344	0.075269	0.148778

Table 2: The largest n-cube in a regular n-simplex of unit edge: the value of Conjecture 4.7, the bound from Lemma F (Corollary 2.6), and the volume bound $\mathrm{vol}(\Delta)^{1/n}$. For $n = 2$ the conjectured value is classical, and for $n = 3$ it is Croft's theorem. For $n \leq ?$ no frame found numerically beats it (Section 6.5).

- $n = 3$ [literature]: Croft [8] determined the largest cube in a regular tetrahedron, and Firsching [9] confirmed it numerically. For edge 1 its side is $1/(1+\frac{2}{3}\sqrt{3}+\frac{1}{2}\sqrt{6}) = 0.29590654$. This is exactly σ_3: with $H = \sqrt{1/2}+\sqrt{2/3}+\sqrt{3/4}$, the side is $1/(\sqrt{2}H) = 1/(1+\sqrt{6}/2+2/\sqrt{3})$. **So Conjecture H holds for the regular tetrahedron, and the face-on-face cube is optimal there.** Croft's argument is global and is the natural template for the facet lemma in dimension 4 (Section 9). We have not read his proof: the paper is behind a paywall.

Asymptotically $\sigma_n \sim 1/(\sqrt{2}\,n)$, while the volume bound is about $e/(\sqrt{2}\,n)$. We found no result on the largest cube in a regular simplex for $n \geq 4$.

Cubes in a tetrahedron in the catalogue. The catalogue's record for two unit cubes in a regular tetrahedron (`cubintet`, $n = 2$) is $2+\sqrt{72\sqrt{2}+102}/6 \approx 4.37945$. Since $\sqrt{72\sqrt{2}+102}/6 = \sqrt{6}/2 + 2/\sqrt{3}$, this is exactly $1/\sigma_3 + 1$, Croft's optimum for one cube plus one. Whether $s_{\mathrm{cubintet}}(2) = 1/\sigma_3 + 1$ is an open lower-bound problem of the same kind as Conjecture 1.1.

5 Routes that fail

This section. Every route to all dimensions that we tried and that does not work, with the reason and, where possible, a counterexample verified at 50–60 digits. None of the counterexamples violates Lemma F: in all of them F_j is far below 1. They refute methods, not the conjecture.
Data. `pivot_counterexamples.json`, `exact_pivot_counterexample_d9.json`, `fplus_counterexample_d8.json`, with the verifiers `verify_pivot_cex.py` and `verify_exact_pivot_cex.py`.

5.1 The unrestricted near-axis step

Idea. Hosono's near-axis step for Lemma F itself, in every dimension. *What fails.* Lemma K needs $\|Q_i\|_1 \leq 2$ for the rows of the polar factor, and a unit vector in $\mathbb{R}^m$ has ℓ^1 norm up to $\sqrt{m} > 2$ for $m \geq 5$. *Repair.* Restrict to rows of norm at most 2 (Theorem 3.1); then $\|Q_i\|_1 \leq 2$ is inherited. The restriction is harmless, since only restricted Lemma F is needed.

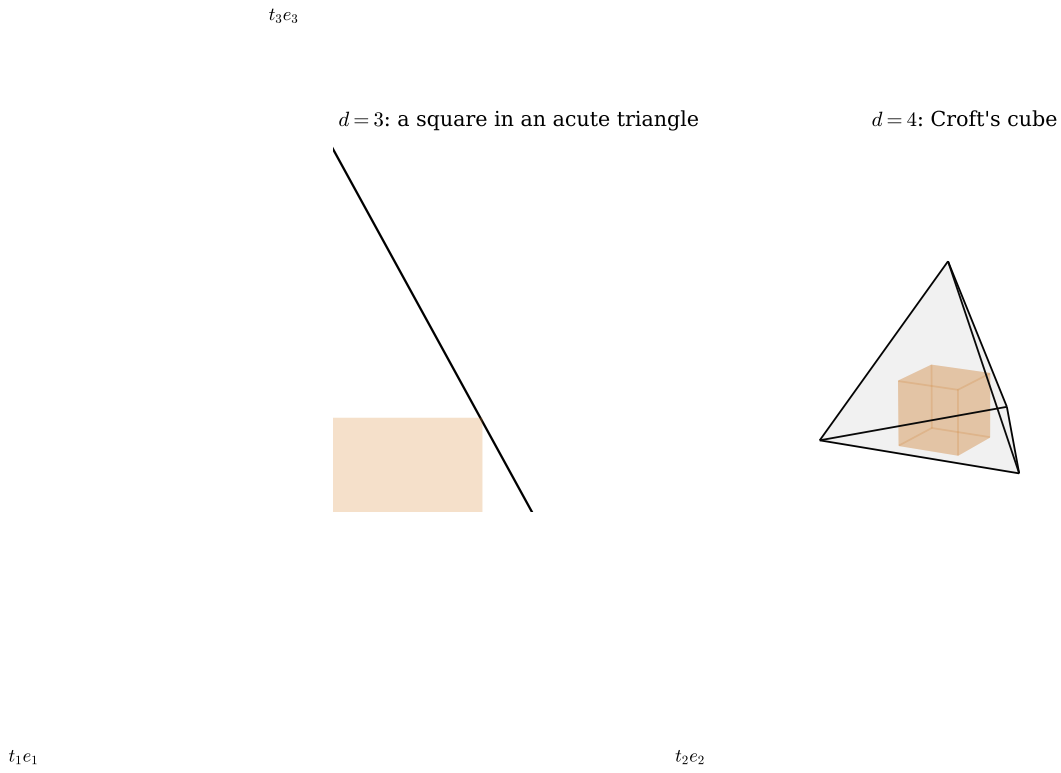


Figure 2: Left: the cube of Theorem 2.5 for $d = 3$: a square in an acute triangle with vertices on the three axes, standing on a side (the Helmert square). Right: the face-on-face cube in a regular tetrahedron, which is the largest cube there by Croft's theorem; its top face is the largest square in the cross-section.

d	ν	c	Ξ	$F_j(Q) - F_j(R)$	$\max_i w_i$
5	0.0051321409	0.5265171977	$-2.5114 \cdot 10^{-4}$	$+9.18 \cdot 10^{-3}$	1.999954
6	0.0029999302	0.5185861643	$-8.9918 \cdot 10^{-4}$	$+4.72 \cdot 10^{-3}$	1.999997

Table 3: Counterexamples to $\Xi \geq 0$ (`pivot_counterexamples.json`, verified with 50-digit polar factors by `verify_pivot_cex.py`; orthogonality error below 10^{-51}).

5.2 Single-pivot inductions

Idea. In the remaining case, pivot at the row p where column j is smallest and the largest entry q of that row. If $\Xi \geq 0$ there ("the pivot conjecture"), Theorems 3.6 and 3.8 prove restricted Lemma F in every dimension. Theorem 3.7 proves it when row p is concentrated.

What fails. $\Xi \geq 0$ is false at spread pivots. Table 3 lists two counterexamples. In both, row p is the unique smallest row of column j, column q holds the unique largest entry of row p, the pivot is admissible, and every row of R and of Q has ℓ^1 norm below 2. Both lie near the equality case $[1] \oplus R'$ of Lemma F, where row p of R' is the flat row $\frac{1}{2}(1,1,1,1)$. There the near step applies anyway, but the pivot inequality does not. The mechanism: rows whose entries have mixed signs relative to row p shrink under the pivot, while the triangle inequality in Theorem 3.6 assumes they grow. Random and optimised tests had never sampled these neighbourhoods.

Inside the remaining case it still fails. Take the direct sum of the 4×4 block containing the flat row and the row $(\frac{\sqrt{3}}{2}, -\frac{1}{2\sqrt{3}}, -\frac{1}{2\sqrt{3}}, -\frac{1}{2\sqrt{3}})$ with a $b \times b$ block carrying column j, $b \geq 5$, so that $S > 2$ and every rest exceeds 1. Reshape the flat row at order $\sqrt{\nu}$ and mix in column j at order ν (`blockrc3.py`). Then every hypothesis of RC4 holds, and in $d = 9, 10$ about 16% of the samples have $\Xi < 0$, with Ξ/ν down to -0.62.

The exact pivot inequality fails too. Even the exact comparison $F_j(R) \le F_j(Q)$, without the triangle inequality, fails in $d = 9$ (`exact_pivot_counterexample_d9.json`, verified at 60 digits by `verify_exact_pivot_cex.py`). The matrix lies strictly inside the remaining case: every row has $w_i < 2$ ($\max w_i - 2 = -1.4 \cdot 10^{-5}$), $\min_i(w_i - x_i) - 1 = 0.087$, $S - 2 = 0.072$, row p is the unique smallest row (gap 10^{-8}) and q its unique largest entry (gap $5 \cdot 10^{-4}$), the pivot is admissible, and every row of Q has norm below 2. Yet $F_j(Q) - F_j(R) = -1.26 \cdot 10^{-8}$, with $F_j(R) = 0.3166$. The effect is first order in $\nu = 1.6 \cdot 10^{-7}$, so it is not rounding.

Conclusion. No rule of the form "pivot at the smallest row of column j, at the largest entry of that row" carries the induction through every dimension. In every counterexample F_j is far below 1, so a proof has to treat spread configurations by a direct bound, or carry a quantitative hypothesis with room below 1.

5.3 Averaging over pivot columns

Idea. Theorem 3.6 needs no maximality: any column $q \neq j$ with $R_{pq} \neq 0$ gives $F_j(R) \le F_j(Q^{(q)}) - a_q \Xi_q$. Such a pivot is admissible when $c_q = |R_{pq}| \ge (w_p - 1)/2$. Average Ξ_q with weights c_q. By row orthogonality, $\sum_q R_{pq} R_{iq} = -\nu y_i$, the terms $-\nu x_i G_i$ combine into $+\nu^2 y_i G_i \ge 0$. The constants combine to $\nu^2(1 - \nu w_p) \ge 0$, and the κ_1-terms are nonnegative for admissible q. In closed form, $\sum_q c_q \Xi_q = \nu^2(1 - 2\nu + F_j(R)) + (\kappa_1\text{-terms})$. So *if every nonzero entry of row p outside column j is an admissible pivot, some admissible q has $\Xi_q \ge 0$.*

What fails. Generic rows have small entries, $0 < c_q < (w_p - 1)/2$. Leaving them out of the average leaves a cross term that, by restricted Lemma F applied to Q, costs only $O(\|\text{small part of row } p\|_2^2)$. We could not absorb this quadratic cost when ν is tiny. In random tests the average over admissible pivots was always nonnegative (`avgq.py`, `avgq2.py`, $d = 5$–10).

A stress test. A heuristic count suggested that many rows with $y_i \approx \nu$, a small positive entry in column q and $G_i \approx 1$ could pile up deficits of order $\sqrt{n}$ against the fixed surplus. Explicit constructions (`construct3.py`, `construct4.py`) show why this does not happen. To share the pivot direction, the parts of those rows outside the pivot block must form a simplex, which forces each to have ℓ^1 norm about 2 and kills the slack G_i. Every feasible structured instance had $\Xi/\nu \ge 1.1$. The protection is a global orthogonality effect, not a row-by-row one, which explains why per-row bounds fail.

5.4 The positive-part strengthening

Idea. Replace $2 - w_i$ by $(2 - w_i)^+$, so that rows of norm above 2 cannot help: $\sum_i x_i (2 - w_i)^+ \le 1$ for all orthogonal R. This would remove the restriction. *What fails.* It is false: an 8×8 orthogonal matrix has $\sum_i x_i (2 - w_i)^+ = 1.0112$ (`fplus_counterexample_d8.json`). So the restriction to rows of norm at most 2 cannot be replaced by truncation. Lemma F itself is not affected, since there the rows of norm above 2 contribute negatively.

5.5 Convex relaxations, and Hosono's spectral bound

Idea. By Corollary 2.4, $\mathrm{Val}_v(C) = \sum_k g(c_k)$ with $g(c) = \sum_i v_i |c_i|$, over orthonormal bases (c_k) of $v^\perp$. Relax the basis to an isotropic measure: any finite measure μ on $v^\perp$ with $\int c c^\top \, d\mu = I - vv^\top$. A lower bound for $\int g \, d\mu$ over all such μ would prove Lemma F, and is a convex problem.

Proposition 5.1 ([proof]). *For uniform v the relaxation fails from $d = 6$ on: the measure with mass $2/d$ on each $(e_a - e_b)/\sqrt{2}$ is isotropic and has $\int g \, d\mu = (d-1)\sqrt{2/d}$, while Lemma F needs $2(\sqrt{d} - 1)$. The difference*

$$q_d = (d-1)\sqrt{2/d} - 2(\sqrt{d} - 1)$$

is 0.0577 for $d = 5$ and -0.0122 for $d = 6$, and decreases from there.

Proof. $\sum_{a<b}(e_a - e_b)(e_a - e_b)^\top = dI - \mathbf{1}\mathbf{1}^\top$, so the weights $2/d$ give $I - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$. Each vector has $g = \sqrt{2}/\sqrt{d}$, and there are $d(d-1)/2$ of them. □

Hosono's spectral bound [5] is $1 - F \geq \sqrt{2}\,\mathrm{tr}\sqrt{\mathrm{diag}(p) - pp^\top} - 2\sum_i \sqrt{p_i} + 2$ for the squared column p. It uses only $\|z\|_1 \geq \sqrt{2}\|z\|_2$ for zero-sum vectors and the triangle inequality for the nuclear norm. At the uniform column it equals q_d exactly, so it is a relaxation of this kind, and it fails for the same reason. This explains why Hosono's proof for $d = 6$ needs missing-edge corrections and 2.8 million certificates, and why no proof that sees only the second moments of the frame can work for $d \geq 6$. A proof must use that the frame is an orthonormal *basis*. The induction of Section 3 does.

5.6 A quantitative band hypothesis

Idea. Carry a stronger hypothesis with room below 1. Numerically $F_j(R) \leq 1 - \varphi(m)$ with $\varphi(m) = (m + \sqrt{1-m^2} - 1)^2$, m the largest entry of column j, $m \geq 1/\sqrt{2}$. This is the value of the plane rotations, and it has no violation in $d = 3$–8 (`bandind.py`). *What fails.* Combined with the near step, which applies the hypothesis to every column of Q, the inductive inequality fails in about 1% of the cases. The losses are in the triangle inequality and in the cross term of Lemma K.

5.7 Local arguments for the facet lemma

Idea. Prove the facet lemma from first- or second-order conditions at a minimiser, as Proposition 4.4 does for $d = 3$. *What fails.* In $d = 4$ (`locmin.py`: 12 random v, 40 local descents each), every v has between 3 and 15 distinct local minima of Val_v. Most instances have local minima with no aligned row: three or four zero entries spread over different rows. Their excess over the Helmert value is sometimes tiny, $7 \cdot 10^{-4}$ or $4 \cdot 10^{-3}$. The global minimum always equals the Helmert value to six digits and has an aligned row, apart from one numerical tie next to an axis-aligned v. So the facet lemma is true numerically in $d = 4$ but is not a local statement. A proof has to compare non-aligned configurations with the Helmert configuration globally. A proof in $d = 4$ would also not lift by itself: the induction needs an aligned minimiser in each dimension, and Lemma F one dimension down says nothing about where the minimiser one dimension up sits.

5.8 Pairwise bounds

Idea. Use only the pairwise inequalities $\langle |C_i|, |C_l| \rangle \geq v_i v_l$ for the rows of a frame, which follow from $\langle C_i, C_l \rangle = -v_i v_l$, together with $\|C_i\|_2^2 = 1 - v_i^2$. *What fails.* The resulting linear relaxation of $\min \mathrm{Val}_v$ is below $2(S-1)$ (`lp.py`). Pairwise information is too weak, in line with Section 5.5.

6 Numerical evidence in dimensions 3 to 32

This section. A systematic search, in the style of Hosono's far-region search, for large values of F_j in dimensions 3 to 32: over all restricted matrices, away from the axes, in the remaining case of the induction, and without the restriction. Then tests of Conjecture H and the facet lemma up to $d =$?.
Scripts and data. `evidence/search.py`, `evidence/frames.py`, `evidence/analysis.py`, `evidence/nearid.py`, `evidence/run_all.py`; results in `evidence/*.jsonl`.

6.1 Method

All searches are batched multi-start optimisations on $SO(d)$ in double precision (PyTorch). A matrix is $R = R_0 \exp(A - A^\top)$ with Adam steps on A. Every 200 steps $R_0 \leftarrow R$ and $A \leftarrow 0$,

d	starts	largest F_j, rows ≤ 2	starts	largest F_j, all of $O(d)$
3	1024	1.0000000000	–	–
4	1024	1.0000000000	–	–
5	1024	1.0000000000	–	–
6	1024	1.0000000000	–	–
7	1024	1.0000000000	–	–
8	1024	1.0000000000	–	–
9	1024	1.0000000000	–	–
10	1024	1.0000000000	–	–
11	512	0.9999999998	–	–
12	512	0.9999999998	–	–
13	512	0.9999999997	–	–
14	512	0.9999999999	–	–
15	512	0.9999999997	–	–
16	512	0.9999999993	–	–
20	256	0.9999999994	–	–
24	256	0.9999999965	–	–
32	256	0.9999999915	–	–

Table 4: Largest F_j found, restricted (rows ≤ 2) and unrestricted, with the number of optimised starts. The column of the maximiser is within 10^{-6} of an axis in every case. No start gave $F_j > 1$.

with a re-orthonormalisation. The absolute values are smoothed to $\sqrt{t^2+\varepsilon^2}-\varepsilon$ with ε annealed from 10^{-2} to 10^{-9}, and the step size from $2\cdot 10^{-2}$ to $6\cdot 10^{-6}$. Constraints enter as quadratic plus linear penalties whose weight grows from 10 to 10^6; there are 6000 steps. The starts mix six families: Haar matrices; signed permutations times exp of a small skew matrix (near the equality set); $[1]\oplus R'$ with R' containing the flat row $\frac{1}{2}(1,1,1,1)$ (the neighbourhoods that refute the pivot inequality); near-uniform columns; sink columns with one large entry; and Hadamard matrices where they exist. At the end every value is recomputed exactly in double precision. A point counts only if it is orthogonal to 10^{-10}, its rows have norm at most $2+10^{-9}$, and its region constraint holds to 10^{-7}.

Validation against Hosono. In $d=4$, Hosono reports that the largest value of F_1 in his far region $c^2 \leq 2/3$ is about 0.845, near the boundary [5]. Our search finds ? there. The exact value of the plane rotation at $c^2=2/3$ is $1-(\sqrt{2/3}+\sqrt{1/3}-1)^2 = 0.844885$.

6.2 The search for a counterexample

Table 4 gives, for each d, the largest F_j found over restricted $O(d)$ and over all of $O(d)$. In every dimension it is 1 up to the optimiser's tolerance, attained at a column within 10^{-6} of a coordinate axis, and no start exceeded 1. The global search therefore only confirms the equality set. The informative numbers are those away from it.

6.3 Away from the axes: Hosono's far region

For $\tau < 1$ let $\Phi_d(\tau)$ be the supremum of F_j over restricted matrices whose column j has every squared entry at most τ. Hosono's far region in dimension d is $\tau_d = 1-1/(d-1)$, the complement of the region where the near-axis step applies in his setting. Table 5 gives our values of $\Phi_d(\tau_d)$ and $\Phi_d(1/2)$, and Fig. 3 the whole profile. Three observations:

- For $\tau \geq 1/2$, the plane rotation at the boundary of the region, $1-\varphi(\sqrt{\tau})$, is the best we found or within ? of it in every dimension. The profile does not depend on d there.
- Below $\tau = 1/2$ the profile flattens: F_j cannot be large when column j is spread. Far from the axes, in the range where only RC-type matrices live, the values stay below 0.0000.

d	τ_d	$\Phi_d(\tau_d)$	$\Phi_d(1/2)$	smallest τ tested: $\Phi_d(\tau)$

Table 5: The supremum $\Phi_d(\tau)$ of F_j over restricted matrices with every squared entry of column j at most τ: Hosono's far region $\tau_d = 1 - 1/(d-1)$, and $\tau = 1/2$. The last column is the plane-rotation value $1 - \varphi(\sqrt{\tau})$ at $\tau = 1/2$, $1 - (\sqrt{2}-1)^2$.

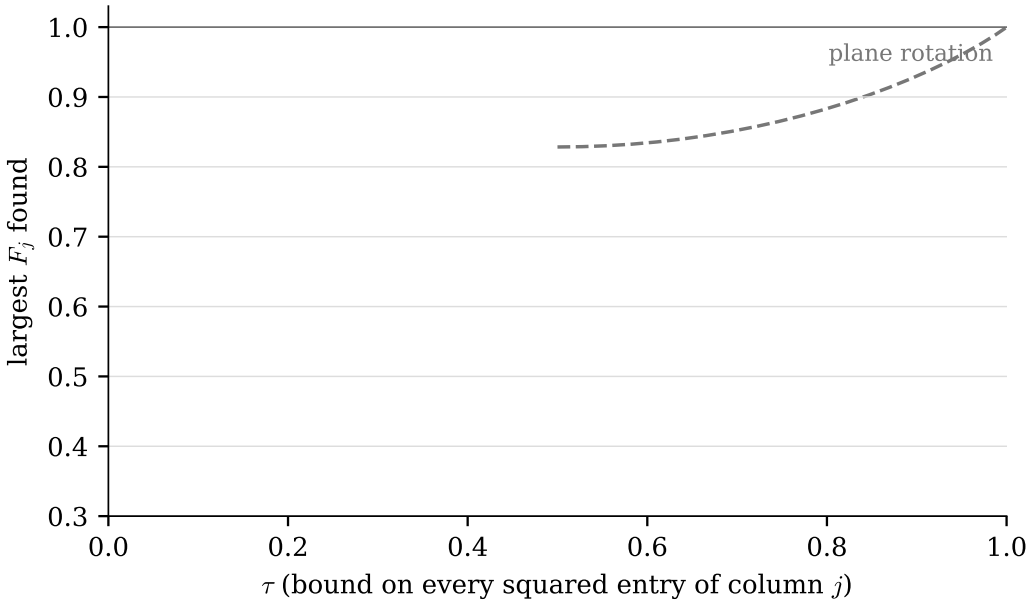


Figure 3: The far-region profile $\Phi_d(\tau)$: the largest F_j found when every squared entry of column j is at most τ, for several dimensions, and the value $1 - (\sqrt{\tau} + \sqrt{1-\tau} - 1)^2$ of the plane rotation at the boundary. Lemma F says every curve stays below 1.

- Hosono's far region τ_d approaches the axis as d grows, and $\Phi_d(\tau_d)$ approaches 1, but only because the region does: at fixed τ the supremum is the same in every dimension.

6.4 The remaining case

Table 6 gives the largest F_j found in the remaining case RC(d) of Theorem 3.4: rows of norm at most 2, $\|x\|_1 \geq 2$, and $w_i - x_i \geq 1$ in every row. This is where a counterexample would have to live, since everything else is reduced to dimension $d-1$ in Lean. The case is empty for $d \leq 4$. In every dimension we tested the supremum is at most 0.9037, and the maximiser sits at the boundary $\|x\|_1 = 2$, where Theorem 3.2 gives $F_j \leq 1$ anyway. RC4(d) $\subseteq$ RC(d), so the same bound holds there.

6.5 Conjecture H and the facet lemma

For each $d \leq ?$ we took 11 vectors v: uniform, three random, two sparse (cubes of Gaussians), three sinks $(c, t, \ldots, t)$ with $c = 0.9, 0.7, 0.5$, a geometric sequence, and two large entries. For each, we minimised Val_v over frames from ? starts, half random and half perturbed Helmert frames, with the same smoothing (`frames.py`). Table 7 summarises. In every case the smallest value found is the Helmert value to within ?, never below it by more than rounding, and the minimiser found has an aligned row in 0 of the 0 cases. For uniform v this is Conjecture 4.7 for $n \leq ? - 1$.

d	starts	largest F_j in RC(d)	$\|x\|_1$	$\min_i(w_i - x_i)$
5	1024	0.729810	2.00015	1.0417
6	1024	0.787949	2.00025	1.0235
7	1024	0.824546	2.00033	1.0144
8	1024	0.850000	2.00042	1.0095
9	1024	0.868924	2.00071	1.0006
10	1024	0.883160	2.00047	1.0007
11	512	0.894715	2.00075	1.0008
12	512	0.903677	2.00108	1.0012

Table 6: Largest F_j found in the remaining case RC(d), with $\|x\|_1$ and $\min_i(w_i - x_i)$ at the maximiser. The bound $F_j \le \|x\|_1 - 1$ of Theorem 3.2 holds everywhere in RC.

d	v tested	starts	$\min(\text{found} - H)$	$\max(\text{found} - H)$	$\min(H - 2(\|v\|_1 - 1))$	aligned

Table 7: Conjecture H: for each d, the smallest and largest difference between the best value found and the Helmert value $H(v)$ over the 11 vectors v, the smallest difference $H(v) - 2(\|v\|_1 - 1)$ (the room in Lemma F), and the number of v whose best frame found has an aligned row.

6.6 Near the equality set

Lemma F is tight on a large set: $F_j(R) = 1$ whenever column j is $\pm e_i$, for any $R = \pm e_i \oplus R'$. Near it, with θ the angle between column j and the nearest axis, the smallest ratio $(1 - F_j)/\theta^2$ that we found is that of a rotation in one coordinate plane, $(\cos\theta + \sin\theta - 1)^2/\theta^2 = 1 - \theta + O(\theta^2)$ (`nearid.py`, $d = 3$–10, $\theta \approx 10^{-3}$; minimum 0.99929). So the inequality is non-degenerate to second order transversally to the equality set, in every dimension tested.

6.7 Earlier tests

The following runs were made during the work and are recorded in the notes `docs/LEMMA_F_ALL_D.md`:

- `cexsearch.py`: a first counterexample search in $d = 7, 9, 12, 16$, with no restricted matrix with $F > 1$;
- `bandind.py`: the band profile $F_j \le 1 - \varphi(m)$ with no violation in $d = 3$–8;
- `helmert.py`, `helmert2.py`, `uniform.py`: local optimisation never went below the Helmert value;
- `star.py`, `rules.py`, `flag.py`, `locmin.py`, `avgq.py`: see Sections 4.3 and 5.7 and Section 5.2;
- `regionW.py`, `rcB.py`: in spread configurations $F_j \lesssim 0.5$.

7 The evidence, collected

Proofs.

- $s_d(2) = 2$ for $d \le 5$ in Lean, and for $d = 6$ by Hosono's computer-assisted proof.
- In every dimension, in Lean: the near step, the off-column step and the concentrated pivot reduce restricted Lemma F in dimension d to dimension $d - 1$ outside the remaining case RC4(d) (Theorem 3.8). The aligned step reduces Lemma F in dimension d to dimension $d - 1$ for every matrix with an aligned row (Theorem 3.9).

File	Lines	Content
`LemmaFr.lean`	200	restricted Lemma F; near step in every dimension; remaining bound; $d \leq 4$; induction step
`OffColumn.lean`	306	off-column step; second induction step
`PivotStep.lean`	647	pivot inequality; concentrated pivot; sign normalisation; RC4 induction step
`FacetStep.lean`	255	value of a column; aligned step; facet lemma; Main Theorem B

Table 8: The Lean files of this paper, on top of those of Paper I (`lean-optimality/PackingOptimality/`).

- The conjecture in every dimension follows from the facet lemma in dimensions $d \geq 6$ (Theorem 3.11, Lean), and the facet lemma follows from Conjecture H (Proposition 4.3).
- The facet lemma holds in $d = 3$ (Corollary 4.5). Conjecture H holds for the regular tetrahedron (Croft) and for the equilateral triangle (classical).
- Elementary facts in every dimension: Lemma F holds on average over the columns (Proposition 2.9). It holds for every orthogonal matrix that is, up to signed permutations, a direct sum of blocks of size at most 5 (at most 6 with Hosono's result), because F_j of a direct sum is F_j of the block containing column j. And it holds whenever $\|\text{column } j\|_1 \leq 2$ and no entry admits the near step (Theorem 3.2).

Structure. The only maximisers found in any dimension are the equality set (columns on an axis). The inequality is non-degenerate to second order transversally to it, with the plane rotations as the worst direction. The extremal family, plane rotations, gives the same profile in every dimension. The Helmert frame has a closed form and a recursive geometric meaning that matches the two known cases.

Numerics. No counterexample in $d \leq 32$. Away from the axes F_j is far below 1, and the far-region profile does not depend on the dimension. In the remaining case, where a counterexample would have to live, $F_j \leq 0.9037$ in every dimension tested. Conjecture H and the facet lemma hold in every test up to $d =?$.

What failed. Every refuted statement is a strengthening or a proof method: the pivot inequality, the exact pivot inequality, the positive-part strengthening, convex relaxations, local arguments. In all the counterexamples F_j is far below 1.

What would change our view. A counterexample must lie in RC4(d) for some $d \geq 7$: rows of norm at most 2, $\|\text{column } j\|_1 > 2$, no entry admitting the near step, and a spread smallest row. Equivalently, it is a frame of $v^\perp$ whose cube in $\Delta_{S-1}(v)$ fits strictly inside, with value below $2(S-1)$. Numerically the remaining case has a lot of room, so such a frame would have to be very special.

8 Formalization

Table 8 lists the files. They build on Paper I's `NearAxis.lean` (Hosono's file, MIT licence), `LemmaF4.lean`, `LemmaF5.lean` and `CentreGeneral.lean`. Every theorem cited is listed in `Axioms.lean` and depends only on `propext`, `Classical.choice` and `Quot.sound`. The repository contains no `sorry`, and its continuous integration builds the project and runs the axiom audit on every change. The Lean links point to commit a58da7c247, the scripts and data to commit test.

Not formalized. Proposition 2.2 (the converse of Theorem 2.1, whose packing step is in Lean). The cube-in-simplex form and its variants (Theorem 2.5 and Proposition 2.8). Propositions 4.1, 4.3 and 4.4 and Corollary 4.5. All numerical statements. The facet lemma itself is a hypothesis of Theorem 3.11: it is stated in Lean (`FacetLemma`) but not proved for $d \geq 3$.

9 Open problems

1. **The facet lemma for** $d \geq 4$ (Definition 3.10). By Theorem 3.11 it is the only missing piece for $d \geq 6$. It needs a global argument (Section 5.7). Croft's proof for the regular tetrahedron is the first thing to study; extending it to the orthocentric tetrahedra $\Delta(v)$, $v \in \mathbb{R}^4$, would give the facet lemma in $d = 4$.
2. **Conjecture H**, and the frame-step inequality (Conjecture 4.6).
3. **A direct bound in the remaining case.** Numerically $F_j \leq 0.9037$ on $\mathrm{RC}(d)$. Any bound below 1 there, with restricted Lemma F in dimension $d-1$, proves dimension d (Theorem 3.4).
4. **The largest cube in a regular simplex** (Conjecture 4.7), open for $n \geq 4$.
5. **Dimension 6 in Lean.** Hosono's proof uses 2.8 million certificates. A Lean proof through Theorem 3.8 needs only the remaining case in dimension 6.
6. **Two cubes in a tetrahedron.** Is the catalogue's record $1/\sigma_3 + 1$ optimal?

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